DATE :
PAGE NO.

⟹ Theme : Online Examination and Result Management System

⟹ Definition & Abstract

The system is a role based digital platform for educational institutions to conduct online examinations, manage courses and enrollments, automate grading and publish results. Core users of system includes admin, faculty staff and student users.

- Based on user type features provided to each user are as follows :

(1) Admin : upload student & faculty data using CSV files. (Authorise access)

(2) faculty : Create Tests, review results through instant auto-grading system, provide feedback for results, Exam scheduling.

(3) Students : Timed interface for taking exams. Instant alert on release of result, get exam feedback; interact with faculty.

Additionally, a common discussion community is provided.

⇒ Team & Role

Product Owner : Sachi Patel (24BCP243)
Scrum Master : Mansi Sharma (24BCP247)
Developer : Daksh Shah (24BCP240)
Mahir Shah (24BCP261)

⇒ Board : Github

⇒ ~~Mo~~Tech stack :

- Backend (Python 3.10+)
  - Web Framwork : FastAPI (Async HTTP server using uvicorn)
  - Database : MySQL
  - Database ORM : SQLAlchemy 2.0
  - Database Driver : aiomysql
  - Authentication & Security
    - ↳ stateless sessions using JWT
    - ↳ Secure password hashing using passlib with bcrypt
  - Data processing : Pandas
  - File uploads : python-multipart
  - Document Generation : reportlab (For producing PDF grade books & certificates
  - Cloud - Storage : google-cloud-storage

- Frontend
  - Structure : Vanilla HTML5
  - Styling : Bootstrap 5.3 for responsive styling and customized Vanilla CSS
  - Interface Elements : Bootstrap Icons
  - Logic : Pure Vanilla JavaScript utilizing asynchronous fetch for all API communication (No frontend framework like react or Vue)